

Internship Project Phase 1 Report (Blue Print)

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Project: E-Commerce Sales Performance Dashboard with Cohort Analysis

External guide:

- 1. Rakshit Shetty – Director, Rooman Technologies**
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1. Introduction

In today's data-driven digital economy, retail businesses generate vast amounts of transactional data through daily sales activities. However, without structured analysis, this data remains underutilized and fails to provide actionable insights. Businesses often struggle to understand customer purchasing behavior, retention patterns, and overall sales performance.

This project focuses on developing an **E-Commerce Sales Performance Dashboard with Cohort Analysis**. The dataset consists of detailed transaction-level records, including customer IDs, invoice details, product information, and timestamps. This project aims to transform raw transactional data into meaningful insights. The analysis will help stakeholders understand customer retention trends, identify revenue drivers, and improve decision-making related to sales and customer engagement.

2. Problem Statement

Retail businesses often lack an integrated analytical framework to effectively monitor sales performance and customer retention. Despite having access to transactional data, organizations face challenges in identifying repeat customer behavior, understanding purchasing patterns, and tracking long-term customer engagement.

The absence of cohort-based analysis limits the ability to evaluate customer lifecycle trends, resulting in missed opportunities for improving retention and maximizing revenue. Therefore, there is a need to develop a system that analyzes transactional data and provides insights into customer retention, sales trends, and business performance.

3. Objectives

The objectives of this project are as follows:

- To analyze **overall sales performance** and revenue trends
- To identify **top-performing products and regions**
- To study **customer behavior and frequency**
- To perform **cohort analysis** to evaluate customer retention over time
- To design and develop an **interactive dashboard using Power BI**
- To generate actionable **insights for improving business performance**

4. Dataset Description

The project utilizes the **E-Commerce Customer Behavior dataset**, which contains detailed transaction-level data from a retail business.

- **Time Period** : April 2018 – March 2019
- **Business Type**: Online Retail.
- **Customer Type**: Individual Customers.
- **Data Nature**: Transaction-level data.

Key Attributes:

- Order ID (unique transaction identifier)
- Category and Sub-category
- Quantity of items purchased
- Order Date
- Total Amount
- Profit
- City and State

This dataset provides sufficient detail for performing **sales analytics, customer behavior analysis, and cohort-based retention analysis**.

5. Scope of the Project

The scope of this project includes the following analytical components:

5.1 Sales Performance Analysis

- Revenue trends over time (monthly and yearly)
- Identification of peak sales periods
- Overall business growth patterns

5.2 Product Analysis

- Identification of top-selling products
- Revenue contribution by product categories
- Product demand distribution

5.3 Customer Analysis

- Total customer count
- New vs returning customers
- Purchase frequency and behavior

5.4 Geographic Analysis

- Country-wise sales distribution
- Identification of key revenue-generating regions

5.5 Cohort Analysis

- Grouping customers based on first purchase month
- Tracking retention over subsequent periods
- Understanding customer lifecycle and churn patterns

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6. What Can Be Done Using This Dataset

The dataset enables the following analytical possibilities:

- Time-series analysis of revenue and Sales
- Identification of customer retention and churn trends
- Geographical Sales comparison
- Creation of cohort-based retention matrices

7. What Will Be Done in This Project

The project will specifically implement the following:

- Data cleaning and preprocessing using Python
- Removal of invalid records such as cancellations and missing values
- Creation of derived features such as revenue and time-based attributes
- Exploratory Data Analysis (EDA) with visualizations
- Cohort analysis to calculate customer retention rates
- Development of a Power BI dashboard with key performance indicators

8. Hypothesis

The project is based on the following assumptions:

1. Repeat customers contribute a significant portion of total revenue
2. Customer retention decreases after the first purchase month
3. A small percentage of products generate the majority of revenue

4. Certain geographic regions contribute more to overall sales

9. Methodology

The project will follow a structured analytical approach, including data collection and loading, cleaning and preprocessing, exploratory analysis, cohort-based customer analysis, and dashboard visualization using Power BI. The focus will be on transforming raw data into meaningful insights for business decision-making.

10. Tools and Technologies

- **Python** (Pandas, Matplotlib, Seaborn)
- **Jupyter Notebook**
- **Power BI** (Dashboard and visualization)

11. Resources Used

The following resources will support the project:

- E-Commerce Customer Behaviour (Kaggle)
- Python documentation and data analysis libraries
- Power BI Desktop
- Online tutorials and reference materials
- ChatGPT

12. Project Plan (2-Day Execution)

Day 1: Data Preparation and Analysis

- Perform data cleaning and preprocessing
- Handle missing values and cancellations
- Create revenue and time-based features
- Conduct exploratory data analysis
- Generate initial insights and visualizations

Day 2: Cohort Analysis and Dashboard Development

- Perform cohort analysis and retention calculation
- Create retention matrix

- Export processed data
- Build Power BI dashboard
- Finalize insights and prepare presentation

13. Expected Outcomes

The project is expected to deliver:

- A fully functional **interactive dashboard**
- A **cohort retention matrix** showing customer behavior
- Clear insights into **sales trends and product performance**
- Identification of **key revenue drivers**
- Actionable recommendations for business improvement

14. Conclusion

This blueprint outlines a structured approach to developing a comprehensive e-commerce analytics solution. By combining sales performance analysis with cohort-based customer retention analysis, the project aims to deliver valuable insights that support data-driven decision-making and business growth.